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Project Name: Device Lending and Access Tracker Application

1. **Introduction and Theme**

For this project I selected the Digital Tools/ Tech Access theme from the Smart Docklands Call for Pilots. This theme focuses on improving access to digital technology for people living, working and studying in Dublin's Docklands. As modern services such as education, employment applications, and government services increasingly reply on digital platforms.

It is important that communities have fair access to devices and digital tools. However, not everyone has access to their own laptop or tablet. Community centres, libraries, and training hubs often provide shared devices for residents to use. The aim of this project is to design and develop a Device Lending and Access Tracker, a java Application that helps organisations manage shared devices and track lending activity. The system allows staff to manage devices and borrowers while ensuring that devices are allocated and returned correctly.

2. **Problem Description**

In the docklands area, many services, jobs and learning opportunities require regular access to computers or tablets. However, not all residents have their own up-to-date devices, and community organisations often need to share laptops or tablets. Now this can be managed with simple sign-out sheets or spreadsheets, which are easy to lose, hard to keep updated and provide little insight into how devices are used.

This can lead to confusion about which devices are available, which ones are currently on loan, and when they are due back. It also makes it difficult for staff to see patterns in demand, such as peak times or the types of devices that are most popular. A digital system can solve these issues by providing a structured way to manage devices, borrowers, and loan records. By tracking this information, staff can quickly see device availability, create loans and manage returns more efficiently.

3. **Proposed Application**

The proposed solution is a Device Lending & Access Tracker, a Java application. The system is designed to support two types of users i.e. Staff and Borrowers. When the application starts, the user chooses to continue as Staff or Borrower. Staff members have administrative over the system, while borrowers can view device availability and their loan history.

Internally, the system represents people and items using classes such as User, Staff, Borrower, Device and Loan. Devices, borrowers and loans are stored in abstract data types, including single linked list, a queue and a stack. The GUI provides form and tables so that staff can perform create, read, update and delete (CRUD) operations on these objects.

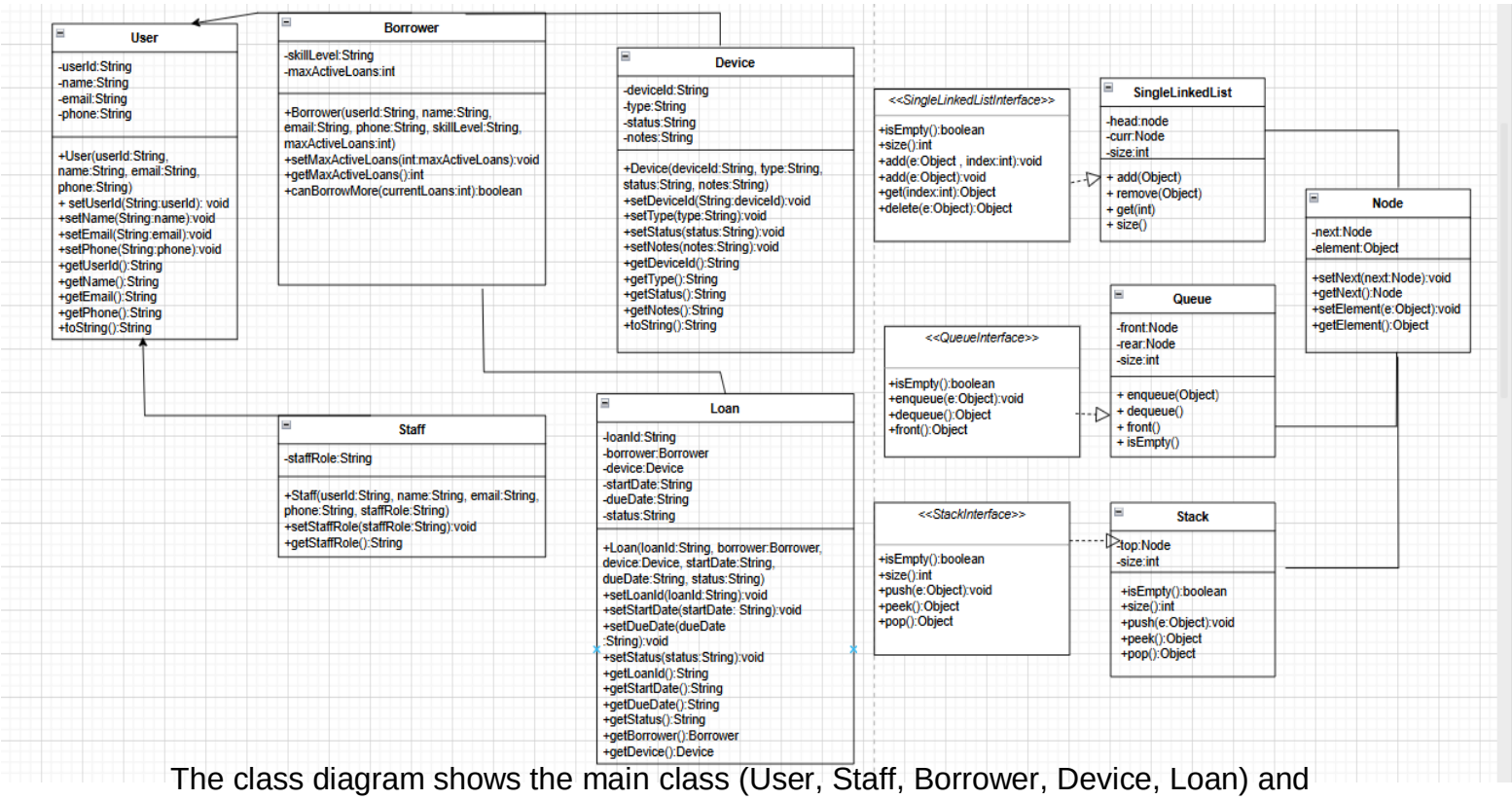
4. **Fit with the Smart Docklands Theme**

This application supports the Digital Tools/ Tech Access theme by helping a Docklands organisation share digital devices in a more organised way. By tracking which devices are available, on loan or under maintenance, staff can make better decisions about how to use limited equipment and identify times when extra devices might be needed. Residents who may not own personal devices can borrow laptops or tablets when needed for

education, job searching, or training. The system also helps staff monitor how devices are used over time. For example, staff can see which types of devices are most popular or when demand for devices increases.

Overall, the application helps improve access to technology while supporting the goal of making Docklands a more digitally inclusive community.

5. **Class Diagram**



The class diagram shows the main class (User, Staff, Borrower, Device, Loan) and the relationships between them. It also includes the abstract data type classes and interfaces such as SingleLinkedList, Queue, Stack, SingleLinkedListInterface, StackInterface, and QueueInterface.

6. **Use of Data Structures and OOP Concepts**

This project demonstrates several object-oriented programming principles and abstract data types.

A single linked list is used to store collections of objects such as devices and borrowers. Each element in the list is stored in a node that references the next node in the list. This structure allows dynamic insertion and removal of elements.

A queue is used to manage active loans. The queue follows the First-In-First-Out(FIFO) principle meaning the earliest loan is processed first. Operations include enqueue(), dequeue(), and front().

A Stack is used to store recently closed or returned loans. The stack follows the Last-In-First-Out(LIFO) principle, operations include push(), pop(), and peek().

Interfaces are used to define the behaviour of each data structure. These include SingleLinkedListInterface, QueueInterface, and StackInterface. By implementing these interfaces, the system ensures that each data structure provides the required operations.

Inheritance- Staff and Borrower extend the User class. Polymorphism- Objects of Staff and Borrower can both be treated as User objects where appropriate. Encapsulation- Class attributes are private and accessed through getter and setter methods. Abstraction – interfaces define behaviour without exposing implementation details.

7. **GitHub Repository**
   The GitHub repository for this project will contain the full NetBeans project including the source code, GUI implementation, and report.

   GitHub Repository:
   https://github.com/Dhruva531/DeviceLendingAndAccessTrackerApp.git